SQL Code

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-- GENDER DISTRIBUTION:
WITH gender_percent AS(
SELECT
        Gender,
        COUNT(Gender) AS TotalCount
FROM Customer_Data
GROUP BY Gender
)
SELECT
        Gender,
        TotalCount,
        ROUND(CAST(TotalCount * 100 AS FLOAT) / SUM(TotalCount) OVER(),1) AS Percentage
FROM gender_percent;

-- STATE DISTRIBUTION:
WITH state_percent AS(
SELECT
        State,
        COUNT(State) AS TotalCount
FROM Customer_Data
GROUP BY State
)
SELECT
        State,
        TotalCount,
        ROUND(CAST(TotalCount * 100 AS FLOAT) / SUM(TotalCount) OVER(),1) AS Percentage
FROM state_percent
ORDER BY Percentage DESC;

-- CONTRACT DISTRIBUTION:
WITH contract_percent AS(
SELECT
        Contract,
        COUNT(Contract) AS TotalCount
FROM Customer_Data
GROUP BY Contract
)
SELECT
        Contract,
        TotalCount,
        ROUND(CAST(TotalCount * 100 AS FLOAT) / SUM(TotalCount) OVER(),1) AS Percentage
FROM contract_percent;

--CUSTOMER STATUS DISTRIBUTION:
WITH customer_status_percent AS(
SELECT
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        Customer_Status,
        COUNT(Customer_Status) AS TotalCount
FROM Customer_Data
GROUP BY Customer_Status
)
SELECT
        Customer_Status,
        TotalCount,
        ROUND(CAST(TotalCount * 100 AS FLOAT) / SUM(TotalCount) OVER(),1) AS Percentage
FROM customer_status_percent;


SELECT
        DISTINCT Churn_Category -- Competitor
FROM Customer_Data;

SELECT
        DISTINCT Churn_Reason --Price too high
FROM Customer_Data;

-- NULL VALUES PRESENT:
-- Value Deal,Multiple lines,Internet type,online security, online backup,Device protection plan,premium support,
-- streaming tv,streaming movies,music,unlimited dara,

SELECT
    DISTINCT Value_Deal  -- DEAL 6
FROM Customer_Data;

SELECT
    DISTINCT Internet_Type -- Cellular
FROM Customer_Data


-- Checking null values:
SELECT
    SUM(CASE WHEN Customer_ID IS NULL THEN 1 ELSE 0 END) AS Customer_ID_Null_Count,

    SUM(CASE WHEN Gender IS NULL THEN 1 ELSE 0 END) AS Gender_Null_Count,

    SUM(CASE WHEN Age IS NULL THEN 1 ELSE 0 END) AS Age_Null_Count,

    SUM(CASE WHEN Married IS NULL THEN 1 ELSE 0 END) AS Married_Null_Count,

    SUM(CASE WHEN State IS NULL THEN 1 ELSE 0 END) AS State_Null_Count,

    SUM(CASE WHEN Number_of_Referrals IS NULL THEN 1 ELSE 0 END) AS Number_of_Referrals_Null_Count,

    SUM(CASE WHEN Tenure_in_Months IS NULL THEN 1 ELSE 0 END) AS Tenure_in_Months_Null_Count,

    SUM(CASE WHEN Value_Deal IS NULL THEN 1 ELSE 0 END) AS Value_Deal_Null_Count,
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    SUM(CASE WHEN Phone_Service IS NULL THEN 1 ELSE 0 END) AS Phone_Service_Null_Count,

    SUM(CASE WHEN Multiple_Lines IS NULL THEN 1 ELSE 0 END) AS Multiple_Lines_Null_Count,

    SUM(CASE WHEN Internet_Service IS NULL THEN 1 ELSE 0 END) AS Internet_Service_Null_Count,

    SUM(CASE WHEN Internet_Type IS NULL THEN 1 ELSE 0 END) AS Internet_Type_Null_Count,

    SUM(CASE WHEN Online_Security IS NULL THEN 1 ELSE 0 END) AS Online_Security_Null_Count,

    SUM(CASE WHEN Online_Backup IS NULL THEN 1 ELSE 0 END) AS Online_Backup_Null_Count,

    SUM(CASE WHEN Device_Protection_Plan IS NULL THEN 1 ELSE 0 END) AS Device_Protection_Plan_Null_Count,

    SUM(CASE WHEN Premium_Support IS NULL THEN 1 ELSE 0 END) AS Premium_Support_Null_Count,

    SUM(CASE WHEN Streaming_TV IS NULL THEN 1 ELSE 0 END) AS Streaming_TV_Null_Count,

    SUM(CASE WHEN Streaming_Movies IS NULL THEN 1 ELSE 0 END) AS Streaming_Movies_Null_Count,

    SUM(CASE WHEN Streaming_Music IS NULL THEN 1 ELSE 0 END) AS Streaming_Music_Null_Count,

    SUM(CASE WHEN Unlimited_Data IS NULL THEN 1 ELSE 0 END) AS Unlimited_Data_Null_Count,

    SUM(CASE WHEN Contract IS NULL THEN 1 ELSE 0 END) AS Contract_Null_Count,

    SUM(CASE WHEN Paperless_Billing IS NULL THEN 1 ELSE 0 END) AS Paperless_Billing_Null_Count,

    SUM(CASE WHEN Payment_Method IS NULL THEN 1 ELSE 0 END) AS Payment_Method_Null_Count,

    SUM(CASE WHEN Monthly_Charge IS NULL THEN 1 ELSE 0 END) AS Monthly_Charge_Null_Count,

    SUM(CASE WHEN Total_Charges IS NULL THEN 1 ELSE 0 END) AS Total_Charges_Null_Count,

    SUM(CASE WHEN Total_Refunds IS NULL THEN 1 ELSE 0 END) AS Total_Refunds_Null_Count,

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    SUM(CASE WHEN Total_Extra_Data_Charges IS NULL THEN 1 ELSE 0 END) AS Total_Extra_Data_Charges_Null_Count,

    SUM(CASE WHEN Total_Long_Distance_Charges IS NULL THEN 1 ELSE 0 END) AS Total_Long_Distance_Charges_Null_Count,

    SUM(CASE WHEN Total_Revenue IS NULL THEN 1 ELSE 0 END) AS Total_Revenue_Null_Count,

    SUM(CASE WHEN Customer_Status IS NULL THEN 1 ELSE 0 END) AS Customer_Status_Null_Count,

    SUM(CASE WHEN Churn_Category IS NULL THEN 1 ELSE 0 END) AS Churn_Category_Null_Count,

    SUM(CASE WHEN Churn_Reason IS NULL THEN 1 ELSE 0 END) AS Churn_Reason_Null_Count

FROM Customer_Data;

SELECT
    Customer_ID,

    Gender,

    Age,

    Married,

    State,

    Number_of_Referrals,

    Tenure_in_Months,

    ISNULL(Value_Deal, 'Deal 6') AS Value_Deal,

    Phone_Service,

    ISNULL(Multiple_Lines, 'No') As Multiple_Lines,

    Internet_Service,

    ISNULL(Internet_Type, 'Cellular') AS Internet_Type,

    ISNULL(Online_Security, 'Yes') AS Online_Security,

    ISNULL(Online_Backup, 'No') AS Online_Backup,

    ISNULL(Device_Protection_Plan, 'No') AS Device_Protection_Plan,

    ISNULL(Premium_Support, 'Yes') AS Premium_Support,

    ISNULL(Streaming_TV, 'No') AS Streaming_TV,

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    ISNULL(Streaming_Movies, 'No') AS Streaming_Movies,

    ISNULL(Streaming_Music, 'No') AS Streaming_Music,

    ISNULL(Unlimited_Data, 'No') AS Unlimited_Data,

    Contract,

    Paperless_Billing,

    Payment_Method,

    Monthly_Charge,

    Total_Charges,

    Total_Refunds,

    Total_Extra_Data_Charges,

    Total_Long_Distance_Charges,

    Total_Revenue,

    Customer_Status,

    ISNULL(Churn_Category, 'Competitor') AS Churn_Category,

    ISNULL(Churn_Reason , 'Price too high') AS Churn_Reason

INTO prod_churn
FROM Customer_Data;


SELECT * FROM prod_churn;

-- CREATE VIEW:
Create View vw_ChurnData as
    select * from prod_Churn where Customer_Status In ('Churned', 'Stayed')

Create View vw_JoinData as
    select * from prod_Churn where Customer_Status = 'Joined'
```